Correction of PN Guidance Law in Frequency-Domain Against Maneuvering Targets

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Abstract

This paper addresses the guidance design for tail-controlled missiles intercepting weaving targets. The key challenge is the varying dynamic characteristics of such missiles: at low altitudes with high dynamic pressure, the autopilot dynamics are essentially minimum-phase; however, at high altitudes, a low-frequency right-half-plane (RHP) zero emerges, degrading performance. We present two frequency-domain design methodologies tailored to these regimes. For the high ω_z (minimum-phase) regime, the design is based on the positive realness framework, which guarantees zero miss distance, finite-time stability, and provides explicit saturation bounds. This leads to the practical Zero-Miss-Distance Proportional Navigation Guidance law, requiring only line-of-sight rate measurement. For the low ω_z (non-minimum phase) regime, where the positive realness condition is fundamentally violated, a frequency-domain shaping approach is employed to directly minimize the miss-distance frequency response peak. The paper clarifies this ω_z-dependent design hierarchy, unifying theoretical rigor with practical necessity for tail-controlled missile guidance.

1 Introduction

Proportional navigation guidance performs poorly against targets executing sustained weaving maneuvers. For tail-controlled missiles, this problem is compounded by altitude-dependent dynamics. At high altitudes, reduced dynamic pressure leads to the appearance of a low-frequency right-half-plane zero in the missile's transfer function, significantly altering the guidance loop's frequency response and degrading the performance of both classical and many advanced guidance laws [1].

This work synthesizes two frequency-domain design paradigms to address the full flight envelope of a tail-controlled missile. When the RHP zero frequency ω_z is sufficiently high (effective minimum-phase behavior), the positive realness framework of Gurfil et al. provides a theoretically optimal solution. When ω_z is low, violating the premises of that framework, we resort to a direct frequency-domain shaping method. The paper reviews these methods, establishing a clear design logic based on the underlying dynamics.

2 Literature Review

The frequency-domain analysis of missile guidance was solidified by Yanushevsky [2], who derived the miss-distance frequency response $P(j\omega)$. For tail-controlled missiles, Zarchan et al. [1] identified the critical problem of a high-altitude RHP zero and proposed numerical compensation.

The positive realness framework, developed by Gurfil, Jodorkovsky, and Guelman [3, 4, 5], establishes conditions for zero miss distance and finite-time stability. This framework leads to the practical ZMD-PNG law [6]. However, its core requirement—a positive real loop function—is incompatible with the phase lag from a low-frequency RHP zero. Thus, a different approach is needed for that regime.

3 Modeling and Problem Formulation

We consider the linearized engagement model with constant closing velocity. The tail-controlled missile dynamics are modeled as:

$$G(s) = \frac{1 - s^2/\omega_z^2}{(1+\tau s)\left(1 + \frac{2\zeta}{\omega_n}s + \frac{s^2}{\omega_n^2}\right)}. \tag{1}$$

The parameter ω_z is not fixed but decreases with increasing altitude and decreasing missile velocity [1]. Consequently:

- For low-altitude/high-speed flight (ω_z large), the term $(1 - s^2/\omega_z^2)$ contributes negligible phase lag in the guidance frequency band, and $G(s)$ can be treated as approximately minimum-phase.
- For high-altitude/low-speed flight (ω_z small), the RHP zero introduces significant non-minimum phase lag, fundamentally altering the system's characteristics.

The guidance design objective is to achieve robust performance against weaving targets across both regimes.

4 Frequency-Domain Analysis and Positive Realness

4.1 Miss-Distance Frequency Response

For a sinusoidal target acceleration $a_T(t) = A_T \sin(\omega t)$, the miss distance in the complex domain is given by [2]:

$$Y(t_f, s) = P(t_f, s)\frac{A_T\omega}{s^2 + \omega^2}, \tag{2}$$

where $P(t_f, s)$ is the transfer function from target acceleration to miss-distance. Here we substitute s with $j\omega$ to denote the frequency characteristic. For the linearized PN loop, $P(t_f, j\omega)$ can be expressed as:

$$P(t_f, j\omega) = \exp\left(N\int_{j\omega}^{\infty}\frac{G(\sigma)}{\sigma}d\sigma\right). \tag{3}$$

The magnitude $|P(t_f, j\omega)|$ determines the miss distance amplitude. For the dynamics in Eq. (1), when $\tau\omega_n \ll 1$, $|P(t_f, j\omega)|$ exhibits a resonant peak at a frequency ω^* approximately given by:

$$\omega^* \approx \omega_n\sqrt{1 - 2\zeta^2}. \tag{4}$$

This is the most dangerous weaving frequency. When ω_z is small, the shape of $|P(t_f, j\omega)|$ changes significantly, often with increased peak magnitude.

4.2 Positive Realness Theorem and Its Implications

The positive realness condition provides a powerful framework for guidance design. A transfer function $H(s)$ is said to be positive real if [4]:

1. $H(s)$ is analytic in $\Re(s) > 0$,
2. $H(s)$ is real for real s,

3. $\Re[H(j\omega)] \geq 0$ for all ω.

Theorem 1 (Positive Realness and Guidance Performance)**.** *For the linearized PN guidance loop with total dynamics $G(s)$ and compensator $K(s)$, let $H(s) = K(s)G(s)/s$. If $H(s)$ is positive real, then:*

1. *The system achieves zero miss distance against any bounded deterministic target maneuver.*

2. *The guidance loop is finite-time globally absolutely asymptotically stable (FT-GAAS).*

3. *The missile acceleration satisfies the bound $\|a_M\|_\infty \leq \frac{N}{N-2}\|a_T\|_\infty$ for $N > 2$.*

Proof. The proof involves three parts: (1) zero miss distance follows from the biproperness condition implied by positive realness [3]; (2) FT-GAAS follows from the circle criterion application [4]; (3) the acceleration bound follows from L^p stability analysis [5]. Details are provided in Appendix A. □

This theorem establishes the design goal for the high-ω_z regime: find $K(s)$ such that $H(s) = K(s)G(s)/s$ is positive real.

4.3 Conditions for Positive Realness

For a rational transfer function $H(s) = b(s)/a(s)$, necessary conditions for positive realness include [3]:

1. $H(s)$ has no zeros or poles in the right half-plane,

2. Any poles on the imaginary axis are simple with positive residues,

3. $\Re[H(j\omega)] \geq 0$ for all ω,

4. The relative degree $r[H(s)] \leq 1$.

For $H(s) = K(s)G(s)/s$ to be positive real, $K(s)G(s)$ must be biproper. If $G(s)$ has relative degree m, then $K(s)$ must have relative degree $-m$, i.e., it must contain m zeros to cancel the poles of $G(s)$. The ideal compensator is:

$$K(s) = \prod_{i=1}^{m}(\tau_{Z_i}s + 1). \tag{5}$$

5 Guidance Design for the High-ω_z Regime

5.1 Compensator Design

When ω_z is large, $G(s)$ in Eq. (1) is treated as minimum-phase with relative degree $m = 3$ (considering the third-order denominator). The ideal compensator from Eq. (5) would be a triple lead network. However, pure differentiation amplifies noise excessively. A practical lead-lag implementation is:

$$K(s) = \frac{\prod_{i=1}^{m}(\tau_{Z_i}s + 1)}{\prod_{i=1}^{m}(\tau_{P_i}s + 1)}, \quad \tau_{P_i} \ll \tau_{Z_i}. \tag{6}$$

The time constants τ_{Z_i} are chosen to provide sufficient phase lead around ω^* to make $\Re[K(j\omega)G(j\omega)/j\omega] \geq 0$ in the critical frequency range.

5.2 Equivalent LOS Rate Implementation

The compensator $K(s)$ can be implemented using the equivalent LOS rate concept [6]. Let $\dot{\lambda}_m$ be the measured LOS rate. Define the equivalent LOS rate as:

$$\dot{\lambda}_{eq}(t) = \hat{\dot{\lambda}} + h_1\hat{\ddot{\lambda}} + h_2\,\hat{\dddot{\lambda}} = \mathbf{h}^T\hat{\mathbf{\Lambda}}, \tag{7}$$

where $\hat{\mathbf{\Lambda}} = [\hat{\dot{\lambda}}, \hat{\ddot{\lambda}}, \hat{\dddot{\lambda}}]^T$ are estimates from the noisy measurement $\dot{\lambda}_m$. The guidance command is:

$$a_c(t) = NV_c\dot{\lambda}_{eq}(t). \tag{8}$$

5.3 State Estimation

The estimates $\hat{\mathbf{\Lambda}}$ are obtained using a Kalman-Bucy filter based on a kinematic model of the LOS rate. The state-space model is:

$$\dot{\mathbf{\Lambda}}(t) = \mathbf{A}\mathbf{\Lambda}(t) + \mathbf{g}w(t), \quad \dot{\lambda}_m(t) = \mathbf{c}^T\mathbf{\Lambda}(t) + v(t), \tag{9}$$

where $w(t)$ and $v(t)$ are process and measurement noises, respectively. The filter equations are:

$$\dot{\hat{\mathbf{\Lambda}}}(t) = \mathbf{A}\hat{\mathbf{\Lambda}}(t) + \mathbf{k}[\dot{\lambda}_m(t) - \mathbf{c}^T\hat{\mathbf{\Lambda}}(t)], \tag{10}$$

$$\mathbf{k} = \mathbf{P}\mathbf{c}/r, \tag{11}$$

$$\mathbf{AP} + \mathbf{PA}^T + \tilde{q}\mathbf{g}\mathbf{g}^T - \mathbf{P}\mathbf{c}\mathbf{c}^T\mathbf{P}/r = 0, \tag{12}$$

where $\mathbf{P}$ is the error covariance, r is the measurement noise variance, and $\tilde{q}$ is the process noise intensity.

The weighting vector $\mathbf{h}$ in Eq. (7) is chosen so that the transfer function from $\dot{\lambda}_m$ to $\dot{\lambda}_{eq}$ approximates $K(s)$ in the frequency band of interest. Detailed derivation of the filter and $\mathbf{h}$ selection is in Appendix B.

5.4 Navigation Ratio Selection

From Theorem 1, to avoid saturation while ensuring zero miss distance, the navigation ratio must satisfy:

$$N \geq \frac{2\mu_0}{\mu_0 - 1}, \quad \mu_0 = \frac{a_{M_{\max}}}{a_{T_{\max}}} > 1. \tag{13}$$

6 Guidance Design for the Low-ω_z Regime

When ω_z is small, the RHP zero makes it impossible to satisfy the positive realness condition over all frequencies. The design goal shifts to directly minimizing the peak of $|P(j\omega)|$.

6.1 Miss-Distance Frequency Response with RHP Zero

For the dynamics in Eq. (1) with small ω_z, the integral in Eq. (3) can be evaluated to obtain an explicit expression for $|P(j\omega)|$ (see Appendix C). The result shows that the RHP zero increases the peak magnitude and may shift ω^*.

6.2 Theoretical Foundation: Optimal Control Formulation

The frequency-domain shaping approach can be derived from an optimal control perspective. Consider the linearized engagement model with state vector $\mathbf{x} = [y, \dot{y}, a_M, \dot{a}_M, \ddot{a}_M]^T$, where y

is the relative displacement normal to LOS. The system dynamics including the RHP zero can be represented as [1]:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}a_c(t) + \mathbf{D}a_T(t), \tag{14}$$

where $\mathbf{A}$ incorporates the tail-controlled dynamics from Eq. (1).

The objective is to minimize miss distance at interception while penalizing excessive control effort. This leads to the quadratic cost function:

$$J = \mathbf{x}^T(t_f)\mathbf{Q}_f\mathbf{x}(t_f) + \int_{t_0}^{t_f} [a_c^2(\tau) + \rho a_T^2(\tau)]d\tau, \tag{15}$$

where $\mathbf{Q}_f$ is a positive semi-definite matrix with $\mathbf{Q}_f(1,1) = 1$ (penalizing final miss distance), and ρ is a weighting parameter.

6.3 Optimal Solution and Time-Varying Gain

The optimal control law minimizing Eq. (15) subject to Eq. (14) is given by the linear-quadratic regulator (LQR) solution:

$$a_c^*(t) = -\mathbf{K}(t)\mathbf{x}(t), \tag{16}$$

where $\mathbf{K}(t)$ is obtained by solving the differential Riccati equation backward in time.

For implementation with only LOS rate measurement available, we need to express Eq. (16) in terms of $\dot{\lambda}$. Using the relationship $y = R\lambda \approx V_c t_{go}\lambda$ and $\dot{y} = V_c(\lambda + t_{go}\dot{\lambda})$ under small-angle assumptions, the optimal control can be reformulated as [2]:

$$a_c^*(t) = N_{eff}(t_{go})V_c\dot{\lambda}(t) + \text{terms involving estimated states.} \tag{17}$$

The time-varying gain $N_{eff}(t_{go})$ emerges from this transformation and is given by:

$$N_{eff}(t_{go}) = \frac{\mathbf{K}(t)\mathbf{T}(t_{go})}{V_c}, \tag{18}$$

where $\mathbf{T}(t_{go})$ is the transformation matrix relating $\mathbf{x}$ to $\dot{\lambda}$.

6.4 Theoretical Justification from Frequency Domain

The connection to frequency-domain shaping comes from analyzing the resulting closed-loop transfer function. The magnitude of the miss-distance frequency response becomes:

$$|P(j\omega)| = \left| \frac{W(j\omega)}{1 + N_{eff}(j\omega)H(j\omega)} \right|, \tag{19}$$

where $H(j\omega) = G(j\omega)/(j\omega)$ and $W(j\omega)$ accounts for the RHP zero effect. The gain N_{eff} is designed to minimize the peak of $|P(j\omega)|$ over frequency, which is equivalent to solving a minimax optimization problem:

$$\min_{N_{eff}(\cdot)} \max_{\omega} |P(j\omega)|. \tag{20}$$

It can be shown that the LQR solution in Eq. (16) provides an approximate solution to this minimax problem, particularly when ρ in Eq. (15) is chosen to balance miss distance reduction against control effort [1, 2].

6.5 Practical Implementation via Gain Scheduling

Since analytical solutions for $N_{eff}(t_{go})$ are intractable for high-order systems with RHP zeros, numerical synthesis is employed:

1. Discretize t_{go} over the expected engagement range.
2. For each t_{go}, solve the LQR problem (or the associated Riccati equation) numerically.
3. Extract $N_{eff}(t_{go})$ from the optimal feedback gains using Eq. (18).
4. Store $N_{eff}(t_{go})$ in a lookup table for onboard implementation.

This approach directly addresses the performance degradation caused by the RHP zero and can be viewed as a form of gain scheduling where the schedule is determined by optimal control theory.

7 Simulation and Comparative Analysis

(Placeholder for simulation discussion) Conceptual simulation studies would demonstrate two key points: 1) In the high-ω_z regime, ZMD-PNG achieves near-zero miss distance where classical PN fails, and outperforms a standard OGL designed for first-order lag. 2) In the low-ω_z regime, the frequency-domain shaping law recovers performance that is severely degraded under both PN and the mismatched OGL, validating the need for the regime-specific approach.

8 Discussion and Conclusion

The guidance design for tail-controlled missiles against weaving targets must account for the varying dynamics across the flight envelope. The parameter ω_z serves as the key discriminant:

- **High-ω_z (Approx. Minimum-Phase):** The positive realness framework provides an optimal guidance solution (ZMD-PNG) with theoretical performance guarantees.
- **Low-ω_z (Non-Minimum Phase):** The frequency-domain shaping approach is necessary to mitigate the detrimental effects of the RHP zero.

This work reviewed these two methodologies, presenting a unified perspective on tail-controlled missile guidance that transitions from a theoretically optimal solution to a necessary practical compensation based on the underlying flight conditions.

A Proof of Theorem 1

This appendix provides the detailed proof of Theorem 1, which connects positive realness to guidance performance.

A.1 Zero Miss Distance Condition

Following [3], the miss distance for any input is zero if and only if the transfer function $G_c(s) = K(s)G(s)$ is biproper. For $H(s) = G_c(s)/s$ to be positive real, it must have relative degree $r[H(s)] \leq 1$. Since $1/s$ has relative degree 1, this implies $r[G_c(s)] = 0$, i.e., $G_c(s)$ is biproper. Hence, positive realness implies zero miss distance.

A.2 Finite-Time Stability

The guidance loop can be represented as a Lur'e problem with time-varying gain $1/t_{go}$. By the circle criterion, if $H(s)$ is positive real, then the system is finite-time globally absolutely asymptotically stable for all $t_{go} > 0$ [4].

A.3 Acceleration Bound

For the system with $H(s) = NG(s)/s$, if $H(s)$ is positive real, then from the small gain theorem, the L^∞ gain from a_T to a_M is bounded by $N/(N-2)$ [5]. This provides the saturation avoidance guarantee.

B State Estimation and Weight Selection

This appendix details the state estimator design and weight vector selection for the ZMD-PNG implementation.

B.1 Kinematic Model

The LOS rate dynamics can be modeled as an exponentially correlated random process. The highest derivative (e.g., LOS jerk) is modeled as:

$$\dddot{\lambda} = -\frac{1}{\tau_E}\ddot{\lambda} + \frac{1}{\tau_E}w(t), \tag{21}$$

where τ_E is the correlation time. This leads to the state-space model in Eq. (9).

B.2 Kalman Filter Derivation

The steady-state Kalman filter gain is obtained by solving the algebraic Riccati equation in Eq. (12). The solution depends on the noise statistics r and $\tilde{q}$, which are tuning parameters.

B.3 Weight Vector Selection

The weight vector $\mathbf{h}$ should be chosen such that the frequency response of $\mathbf{h}^T(s\mathbf{I} - \mathbf{A}_f)^{-1}\mathbf{k}$ approximates the ideal compensator $K(s)$ in Eq. (5) over the guidance bandwidth, where $\mathbf{A}_f = \mathbf{A} - \mathbf{k}\mathbf{c}^T$. This can be formulated as a frequency-domain fitting problem.

C Miss-Distance Calculation for Non-Minimum Phase Systems

This appendix derives the miss-distance frequency response for systems with RHP zeros.

C.1 Evaluation of the Integral

For $G(s)$ in Eq. (1), the integral in Eq. (3) can be decomposed using partial fractions. The RHP zero contributes a term:

$$\int_{j\omega}^{\infty} \frac{1 - \sigma^2/\omega_z^2}{\sigma} d\sigma = \ln\left(\frac{j\omega}{\omega_z}\right) + \text{terms from other poles.} \tag{22}$$

This leads to additional phase and magnitude contributions that increase $|P(j\omega)|$ when ω_z is small.

D Gain Synthesis for Frequency-Domain Shaping

This appendix details the numerical procedure for synthesizing $N_{eff}(t_{go})$ and provides the theoretical justification.

D.1 Optimal Control Formulation with RHP Zero

Consider the state-space representation of the tail-controlled missile dynamics including the RHP zero:

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}a_c + \mathbf{D}a_T, \\ y &= \mathbf{C}\mathbf{x}, \end{aligned} \tag{23}$$

where $\mathbf{x}$ includes relative position, velocity, missile acceleration states, and possibly target maneuver states if estimated.

The cost function in Eq. (15) balances final miss distance against control effort and target maneuver magnitude. The optimal control law is:

$$a_c^*(t) = -\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t)\mathbf{x}(t), \tag{24}$$

where $\mathbf{P}(t)$ satisfies the differential Riccati equation:

$$-\dot{\mathbf{P}}(t) = \mathbf{A}^T\mathbf{P}(t) + \mathbf{P}(t)\mathbf{A} - \mathbf{P}(t)\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t) + \mathbf{Q}, \tag{25}$$

with boundary condition $\mathbf{P}(t_f) = \mathbf{Q}_f$.

D.2 Transformation to LOS Rate Formulation

Under the linearized geometry assumptions, we have:

$$\dot{\lambda} = \frac{\dot{y}}{V_c t_{go}} - \frac{y}{V_c t_{go}^2}. \tag{26}$$

The state vector can be expressed in terms of $\dot{\lambda}$ and other states through a linear transformation $\mathbf{x} = \mathbf{T}(t_{go})\mathbf{z}$, where $\mathbf{z} = [\dot{\lambda}, \xi_1, \xi_2, \ldots]^T$ contains $\dot{\lambda}$ and other unmeasured states. Substituting into the optimal control law yields:

$$a_c^*(t) = -\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t)\mathbf{T}(t_{go})\mathbf{z}(t). \tag{27}$$

If we approximate $\mathbf{z}(t)$ by $[\dot{\lambda}(t), 0, \ldots, 0]^T$ (i.e., neglecting the unmeasured states or assuming they are small), we obtain:

$$a_c^*(t) \approx N_{eff}(t_{go})V_c\dot{\lambda}(t), \tag{28}$$

where

$$N_{eff}(t_{go}) = -\frac{1}{V_c}[\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}(t)\mathbf{T}(t_{go})]_1, \tag{29}$$

with $[\cdot]_1$ denoting the first element corresponding to $\dot{\lambda}$.

D.3 Connection to Frequency-Domain Shaping

The closed-loop transfer function from target acceleration to miss distance with this control law is:

$$P(s) = \frac{e^{-N_{eff}\int_s^\infty \frac{G(\sigma)}{\sigma}d\sigma}}{s^2}. \tag{30}$$

For sinusoidal target maneuvers $a_T(t) = A_T\sin(\omega t)$, the miss distance magnitude is $|P(j\omega)|A_T/\omega^2$. The gain N_{eff} is chosen to minimize the worst-case (maximum) value of $|P(j\omega)|$ over ω, which corresponds to the minimax problem in Eq. (20).

The LQR formulation provides a computationally tractable method to approximate this solution. The weighting parameter ρ in Eq. (15) controls the trade-off: smaller ρ leads to more aggressive control (lower miss distance but higher N_{eff}), while larger ρ reduces control effort at the cost of increased miss distance.

D.4 Numerical Implementation

The synthesis procedure follows these steps:

1. Discretize time-to-go from $t_{go}^{\min}$ to $t_{go}^{\max}$ with step Δt_{go}.
2. For each t_{go} value, compute $\mathbf{P}(t)$ by solving the Riccati equation backward from t_f to t_0.
3. Compute $N_{eff}(t_{go})$ using the transformation above.
4. For real-time implementation, store (t_{go}, N_{eff}) pairs in a lookup table.
5. During flight, interpolate N_{eff} based on estimated t_{go}.

This method, while computationally intensive offline, yields a practical guidance law that accounts for the detrimental effects of the RHP zero.

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